WEEK 7 · LECTURE L1 · UNIT III

The Analytics Ladder

Four kinds of question, from what happened to what we should do, and where HR realistically stands.

R K Singh · Senior Professor, Department of Commerce · Delhi School of Economics, University of Delhi
DSEH001 · Digital Transformation in Human Resource Systems · M.Com. PGCF 2025 · Exam preparation edition

AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Distinguish descriptive, diagnostic, predictive and prescriptive analytics with HR questions.
- Explain why prediction needs reliable description underneath it.
- Distinguish leading from lagging indicators.
- Apply Davenport et al., Marler and Boudreau, Rasmussen and Ulrich, and Shmueli to HR analytics.

COURSE OUTCOMES

CLO 3 (people analytics) and CLO 4 (metrics and evidence).

THE LADDER

Four rungs of analytic questions

Rung	The question	An HR example
Descriptive	What happened?	How many employees left last year, by unit?
Diagnostic	Why did it happen?	Do employees under weakly rated managers leave more often?
Predictive	What is likely to happen?	Who is likely to leave next quarter?
Prescriptive	What should we do?	Which managers should be coached first?

EXAM LENS

Do not confuse this ladder with Bersin's maturity levels: this one grades the kind of question; Bersin grades an organization's capability.

WHERE HR SITS

Prediction resting on a foundation that does not exist

The common pattern

Organizations buy predictive flight-risk models while their basic headcount and exit records are still unreliable.

The consequence

Prediction without clean description cannot be checked; errors go unnoticed and can harm the people flagged.

The rule

Climb the ladder in order. An organization that cannot count its exits reliably should not be predicting them.

Act before the loss

Lagging

Records the outcome after it happens: attrition counts people already gone.

Leading

Moves before the outcome: engagement and manager quality often shift before exits rise.

The test

Show that the leading indicator actually comes before the outcome (temporality, Week 3) before relying on it.

THEORY IN DEPTH · DAVENPORT, HARRIS AND SHAPIRO (2010)

Leading firms treat talent decisions as investments

THE CORE IDEA

Talent analytics runs from simple facts to optimization of the talent supply chain.

HOW IT WORKS

Six uses, from simple to advanced: human capital facts, analytical HR, human capital investment analysis, workforce forecasts, talent value models, talent supply chain.

LIMITS AND CRITIQUE

Written from case studies of large, data-rich firms such as Google and Harrah's.

IN PRACTICE

Most Indian HR functions are still at the first use, human capital facts; that is the honest starting point.

EXAM LENS

Locate an organization on the six uses, then argue the next realistic step.

THEORY IN DEPTH · MARLER AND BOUDREAU (2017)

The evidence for HR analytics is thinner than the hype

THE CORE IDEA

An evidence-based review found relatively few rigorous peer-reviewed tests of whether HR analytics improves results.

HOW IT WORKS

Value depends on linking HR data to business outcomes and on HR's own analytical capability.

LIMITS AND CRITIQUE

Published evidence lags practice; absence of evidence is not evidence of no effect.

IN PRACTICE

Before buying more analytics tools, an HR function needs people who can frame a question.

EXAM LENS

Their definition of HR analytics is precise and examinable; quote its key elements.

THEORY IN DEPTH · RASMUSSEN AND ULRICH (2015)

Analytics should start with a business problem

THE CORE IDEA

HR analytics risks becoming a management fad if it starts from available data rather than decisions.

HOW IT WORKS

Begin with a business challenge, then find the data and method that answer it.

LIMITS AND CRITIQUE

Some discovery does come from exploring data; the warning is against exploration with no purpose.

IN PRACTICE

Start from "why are frontline sites losing people?", not from "what can our dashboard show?"

EXAM LENS

Use this to critique dashboards-first analytics programs.

THEORY IN DEPTH · SHMUELI (2010)

To explain or to predict: two different goals

THE CORE IDEA

Models built to explain causes and models built to predict outcomes are judged differently.

HOW IT WORKS

Prediction rewards accuracy on new data; explanation needs variables that truly cause the outcome.

LIMITS AND CRITIQUE

Many studies blur the two and claim causes from predictive accuracy.

IN PRACTICE

A flight-risk model can predict well and still give managers no idea what to change.

EXAM LENS

Distinguishing explanation from prediction is a mark of analytical rigour in any people analytics answer.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"Most HR functions are trying to predict before they can describe." Evaluate this claim using the analytics ladder and the evidence on HR analytics. **20 marks**

A strong answer: defines the four rungs; uses Marler and Boudreau and Rasmussen and Ulrich; applies Shmueli; argues for sequencing.

Distinguish explanatory from predictive models, and discuss the implications for flight-risk analytics in HR. **20 marks**

A strong answer: explains Shmueli's distinction; shows why accuracy does not imply actionable causes; addresses harms of false flags.

Distinguish leading from lagging indicators with HR examples. **10 marks**

A strong answer: gives correct examples and the temporality test.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **Know which rung your question is on before you answer it**
2. **Prediction needs a reliable description underneath it**
3. **Leading indicators let you act before the loss**

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Marler, J. H., & Boudreau, J. W. (2017). An evidence-based review of HR analytics. *IJHRM, 28*(1), 3–26.
- Fitz-enz, J., & Mattox, J. R., II (2014). *Predictive Analytics for Human Resources*, Ch 1 (Where's the Value?) and Ch 2 (Getting Started).

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Davenport, T. H., Harris, J., & Shapiro, J. (2010). Competing on talent analytics. *Harvard Business Review, 88*(10), 52–58.

Fitz-enz, J., & Mattox, J. R., II (2014). *Predictive analytics for human resources*. Wiley.

Marler, J. H., & Boudreau, J. W. (2017). An evidence-based review of HR analytics. *The International Journal of Human Resource Management, 28*(1), 3–26. https://doi.org/10.1080/09585192.2016.1244699

Rasmussen, T., & Ulrich, D. (2015). Learning from practice: How HR analytics avoids being a management fad. *Organizational Dynamics, 44*(3), 236–242. https://doi.org/10.1016/j.orgdyn.2015.05.008

Shmueli, G. (2010). To explain or to predict? *Statistical Science, 25*(3), 289–310. https://doi.org/10.1214/10-STS330